The Islamia University of Bahawalpur
Department of Information Technology

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E-Commerce Application Development

Chapter
21

SEO and Digital Marketing

Part VI — Reach, Law and Practice

Taught in	Week 15
Laboratory	Load test with cache on and off; crawlability measured
Assessment	**Assignment 2 due**

This is a class extract. It is *Chapter 21* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

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Chapter 21

SEO and Digital Marketing

Learning Outcomes

By the end of this chapter you will be able to:

- **Distinguish** the parts of search optimisation you control and can check from the parts you cannot.
- **Measure** a shop's crawlability: URLs, titles, descriptions, canonical links, sitemap and first byte.
- **Explain** why readable URLs are two halves, and **apply** both.
- **Diagnose** a configuration change that makes a shop worse while appearing to improve it.
- **Prioritise** marketing work by what is checkable rather than by what is advised.

Before You Start

Chapter 16: `AllowOverride None`, and why `.htaccess` is never read on this stack. Chapter 8: page weight and time to first byte. Chapter 6: the catalogue tables behind the pages.

Key Terms in This Chapter

crawler • indexing • readable URL • URL rewriting • canonical link • meta description • title tag • duplicate content • sitemap • robots.txt • faceted navigation • time to first byte • structured data

21.1 What Is Checkable, and What Is Not

Most writing about search optimisation concerns things nobody outside the search engine can verify: how a ranking is computed, how a change affected it, what the algorithm rewards this month. This chapter does not go there, because a course that cannot check its claims has no business making them.

The question this chapter asks instead

Not "how do I rank higher", but: **can a crawler read these pages, tell them apart, and find all of them?**

Every part of that is checkable from outside, in about a minute, with the shop in front of you. And a shop that fails it cannot rank at all — so it is the part worth doing first regardless of what you believe about the rest.

21.2 Readable URLs Are Two Halves

Readable URL, URL rewriting

A **readable URL** identifies a page by words rather than by parameters. This course's demonstration category is reached at

`/en-gb/catalog/desktops`

rather than at

`index.php?route=product/category&path=20`

It needs two independent things.

The **application half** generates them: the shop looks each link up in a table of keywords and writes the readable form into its HTML.

The **server half** resolves them: the web server turns the readable path back into the parameters the application understands, *before* the application sees the request.

Either half alone is useless. The second alone leaves the links unchanged. The first alone is *worse than nothing.*

Measured: Each half on its own, and both together

From `code/m21_seo.py -fix`. The shop ships with **78 prepared keywords** in `oc_seo_url` and `config_seo_url` set to **0**.

Configuration	Readable?	HTTP	A link the page emits
setting off, rewrite off	no	200	`index.php?route=...`
setting ON, rewrite off	yes	**404**	`/en-gb/catalog/desktops`
setting ON, rewrite ON	yes	200	`/en-gb/catalog/desktops`

The middle row is the finding. Switching on the setting that every guide tells you to switch on rewrites every link in the shop into a readable one — and **every one of them returns 404**. The home page still works, because its own address is unchanged; everything reachable from it is gone.

And the cause is Chapter 16's. The rewrite ships in `.htaccess.txt`, and `AllowOverride None` means Apache never reads `.htaccess` at all. One misconfiguration, found while looking for an exposed error log, silently disables an unrelated feature two chapters later.

The application half is faultless. Fetching `index.php?_route_=htc-touch-hd` by hand returns 200 — the lookup, the keyword table and the routing all work. Only the server was never told.

A change that makes a shop worse while appearing to improve it

Consider what the shopkeeper sees after ticking that box. The admin panel reports success. The home page loads. The URLs in the address bar are now pretty, which is visibly what was asked for. Nothing in any log is unusual, because a 404 is a perfectly ordinary response.

What has actually happened is that **every product and category page in the shop has stopped existing**, and the first person to find out will be a customer or a crawler.

The habit this argues for is one step long: after any configuration change, follow a link. Not the page you changed — a link from it. The whole of this failure is visible in one click and invisible in any amount of looking at the setting.

21.3 Can a Crawler Tell the Pages Apart?

Measured: Titles and descriptions across the catalogue

From `code/m21_seo.py -meta`, over all 19 product pages.

Product pages checked	19	
Distinct titles	**19**	no duplicates — good
Duplicate titles	0	
Pages with no meta description	**19**	**all of them**
Distinct descriptions	0	

Titles are fine. Each is the product's name, and because product names differ, so do the titles. That is the one thing the platform gets right here, and it gets it right by accident rather than by design — a shop selling "Blue T-Shirt" in four sizes as four products would have four identical titles.

Not one page has a meta description. The field exists in `oc_product_description`; the demonstration data leaves it empty, and the platform writes nothing in its place. A search engine will then compose a description from whatever text it finds on the page, which for a product page is usually the navigation.

And no page carries a canonical link — not the home page, not a category, not a product. So the same product reached through two categories is two URLs with identical content, and nothing tells a crawler which one is the real one.

Measured: What a crawler is told, and how fast

From `code/m21_seo.py -crawl`.

`robots.txt`	HTTP 200	16 `Disallow` rules
… names a sitemap?	**no**	
`/sitemap.xml`	**HTTP 404**	
A sitemap extension	**HTTP 404**	none ships

	Canonical?	**First byte (median)**
Home	no	114 ms
Category	no	133 ms
Product	no	110 ms

The `robots.txt` is good work. Its sixteen rules block sorted, paged and filtered variants of listing pages — `?sort=`, `&page=`, `?filter_name=` — which is exactly the faceted navigation that otherwise generates thousands of near-identical URLs from twenty products. Somebody thought about this.

There is no sitemap of any kind, and `robots.txt` does not name one. A crawler therefore finds pages only by following links, so anything more than a few hops from the home page is, in practice, not there. For a catalogue of nineteen products this does not matter; at Chapter 6's fifty thousand it is most of the shop.

First byte is 110–133 ms, which is the server's own thinking time and is respectable. Chapter 8 showed that the other 600 KB is what the customer actually waits for — so this number, the one most often quoted, is the smallest part of the experience.

21.4 The Order to Work In

Six things, each of which you can verify yourself

1. **Make the pages reachable.** Both halves of readable URLs, or neither. Then follow a link.
2. **Publish a sitemap** and name it in `robots.txt`. The platform ships none, so this is a module (Chapter 19) — a query over products and categories emitting XML.
3. **Give every page a canonical link.** One template override (Chapter 8), and it ends the duplicate-content question for the whole shop.
4. **Write meta descriptions**, starting with the twenty products that sell. It is data entry, it is nobody's favourite afternoon, and it is the only one of these six a search engine shows to a human.
5. **Keep `robots.txt` blocking the facets.** It already does; the risk is somebody replacing it.
6. **Then look at speed** — and from Chapter 8's measurements, at page weight rather than at first byte.

Everything above is checkable in your own shop, today, with no access to anybody's analytics. Advice that cannot be checked that way goes after all six.

On the marketing half of this chapter's title

The 2025 curriculum renames this course *Digital Marketing & E-Commerce*, and a handout traced to the 2017 outline should say what it is not covering. Campaign strategy, content marketing, social advertising and attribution modelling are a different subject with different methods, and three weeks of this course cannot do them justice.

What this course contributes to them is narrower and more durable: **a shop that can be crawled, a coupon code per channel (Chapter 12) so that a campaign is measurable at all, and the habit of asking what the denominator is (Chapter 17) before believing a rate.** Without those three, a marketing budget is being spent on a shop that cannot report what it bought.

Common Pitfalls

- **Switching on readable URLs without the rewrite.** Every link in the shop becomes a 404 and nothing reports it.
- **Changing a setting and checking the page you changed** instead of following a link from it.
- **Assuming `.htaccess` does anything on this stack.**
- **Leaving meta descriptions empty** and letting the engine write them from your navigation.
- **Shipping without canonical links**, so that one product at four URLs competes with itself.
- **Removing the `robots.txt` facet rules** to "let everything be indexed".
- **Optimising first byte** when Chapter 8 measured 600 KB of assets after it.
- **Believing any claim about ranking that you cannot test.**

Try It Yourself: Make your shop findable, and break it first

```
# --- 1. Look at a link before changing anything ---------------------
cd code
```

```
curl -s "http://localhost:8090/index.php?route=product/category&path=20" \
  | grep -o 'href="http://localhost:8090/[^"]*"' | head -3

# --- 2. Turn on readable URLs the way every guide says --------------
# Admin: System > Settings > Edit > Server > Use SEO URLs = Yes.

# --- 3. Now FOLLOW A LINK -------------------------------------------
# Click any category. Write down what happened before reading on.

# --- 4. Measure all three configurations ----------------------------
python m21_seo.py --fix

# --- 5. Explain the middle row --------------------------------------
# Name the chapter that already measured the cause.

# --- 6. Audit what a crawler can tell apart -------------------------
python m21_seo.py --meta
python m21_seo.py --crawl

# --- 7. Fix the cheapest finding ------------------------------------
# Add a canonical link to the product template in your child theme
# (Chapter 8). One line. Re-run --crawl and confirm it appears.

# --- 8. Write the sitemap module ------------------------------------
# Chapter 19's structure: one controller, one query over enabled
# products and categories, XML out. Then name it in robots.txt.

# --- 9. Write ten meta descriptions ---------------------------------
# For ten real products. Then say honestly how long 50,000 would
# take, and what you would do instead.

# --- 10. Restore ----------------------------------------------------
python m21_seo.py --restore
```

What to notice. Step 3. Almost everybody ticks the box, sees a pretty URL in the address bar, and concludes it worked. The entire shop is broken at that moment, and one click shows it.

Checkpoint — Test Yourself

1. A client turns on readable URLs and reports that "the site looks the same but customers say pages are missing". Explain precisely what has happened, name the two halves and which one is absent, give the command that demonstrates it, and say why nothing appeared in any log.
2. All 19 product pages have distinct titles and none has a meta description. Say which of those two facts is luck and which is design, what happens on the pages with no description, and what you would do first for a catalogue of 50,000.
3. This shop has no sitemap and no canonical links. Explain what each absence costs, say which matters more for a 19-product shop and which for a 50,000-product shop, and justify the difference.

Chapter Summary

- Ask what is checkable: can a crawler read the pages, tell them apart, and find all of them? A shop failing that cannot rank whatever else is true.
- Readable URLs are two halves — the application generating them and the server resolving them. Either alone is useless and the first alone is harmful.

- Measured: with the setting on and the rewrite off, every link the shop emits returns **404**, while the home page and `index.php?_route_=...` work perfectly. The cause is Chapter 16's `AllowOverride None`.
- After any configuration change, follow a link. The whole of that failure is visible in one click and invisible in any amount of inspecting the setting.
- Measured: 19 distinct titles and 0 duplicates — by luck, because product names differ — and **19 pages with no meta description at all**, and no canonical link on any page.
- Measured: `robots.txt` blocks the faceted-navigation parameters, which is good work; there is no sitemap of any kind and nothing names one. First byte is 110–133 ms, which Chapter 8 showed is the smallest part of what the customer waits for.
- Work in order: reachable, sitemap, canonical, descriptions, keep the facet rules, then speed — and page weight before first byte.
- This course's contribution to marketing is a shop that can be crawled, a code per channel so campaigns are measurable, and always asking what the denominator is.

Review Questions

1. **(MCQ)** With `config_seo_url` on and no server rewrite, a link the category page emits returns: (a) 200; (b) 301; (c) 404; (d) 500.
2. **(MCQ)** How many of the 19 product pages carried a meta description? (a) all 19; (b) about half; (c) one; (d) none.
3. **(Short)** Explain what a canonical link is for, with an example of two URLs in this shop that need one.
4. **(Short)** Why does `robots.txt` block `?sort=` and `?filter_name=`, and what would happen if it did not?
5. **(Short)** Give the one-step habit this chapter argues for after any configuration change, and the failure it would have caught.
6. **(Applied)** You inherit a shop with 8,000 products, no sitemap, no canonical links and meta descriptions on 200 of them. Set out your plan in order, with the effort and the expected benefit of each step, and say which steps you would automate and how.
7. **(Applied)** Specify the sitemap module this platform lacks: what it queries, how it handles a 50,000-product catalogue within the sitemap size limits, how often it is regenerated, how it is named in `robots.txt`, and how you would verify it works without waiting for a search engine.